
Heard: A Korean Long-Term Memory Benchmark and On-Device Retrieval Pipeline for Solo-Business Monologue

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🔗 Code: <https://github.com/cykim05/heard>

🗂️ Dataset: <https://huggingface.co/datasets/chanyoungkim/heard-bench>

Abstract

Heard is an on-device Korean assistant that mirrors a user’s past self back at decision moments, aimed at Korea’s solo-business owners who process nightly self-talk alone. Evaluating its memory requires a long-term-memory benchmark that is simultaneously Korean, monologue rather than dialogue, and grounded in solo-business decisions—a combination no existing benchmark (LongMemEval, PerLTQA, LoCoMo) satisfies. We construct heard-bench, a 270-item benchmark across three tracks (en_subset, ko_translated, ko_native) built from a 2,046-utterance synthetic corpus over three personas, adversarially filtered against no-memory baselines and four-gate auto-validated. We run eleven system-under-test (SUT) configurations spanning four model families (Kakao Kanana 1.5, Alibaba Qwen 2.5, NAVER HyperCLOVA-X SEED, and Korean-tuned Llama-3 derivatives B blossom and Open-Ko), three parameter scales (1.5–8 B), and a 4-bit NF4 axis—14,850 generations and 2,112 pairwise verdicts. Three findings are robust: dense retrieval lifts contains-pass on ko_native for every configuration (mean +11.8 pp, no negative deltas); a reflective response policy dominates a generic advisory baseline on every rubric (88.1% and 88.3% on the two “mirror, do not advise” rubrics); and the ordering en_subset ≤ ko_translated < ko_native holds for ten of eleven SUTs. A v0.3 validation pass then closes three self-named limitations: (i) a position-bias audit shows the two headline rubrics are also the position-stable ones; (ii) a BM25 baseline matches dense gold recall yet diverges by scale, locating the bottleneck in the reader rather than the retriever; and (iii) an “answer-first” guardrail predicted to close the retrieval > oracle regression fails to, refuting the prompt-artifact hypothesis in its strong form.

1 Introduction

1.1 Background and the benchmark gap

Heard is an on-device Korean LLM assistant whose architecture rests on three pillars. **MIC** handles always-off, tap-to-talk speech-to-text (implemented in v0.1 but not evaluated here, so the memory-pipeline question is isolated from transcription error; all evaluation operates on gold utterance text). **NODE** is the domain-specific typed memory built over a user’s past utterances. **MIRROR** produces reflective responses that quote the user’s past words, avoid imperatives, and end with an open question rather than issuing advice. The product targets Korea’s roughly 1.4M-member solo-business community, where nightly self-talk arrives from people with no coworkers and no one to check in

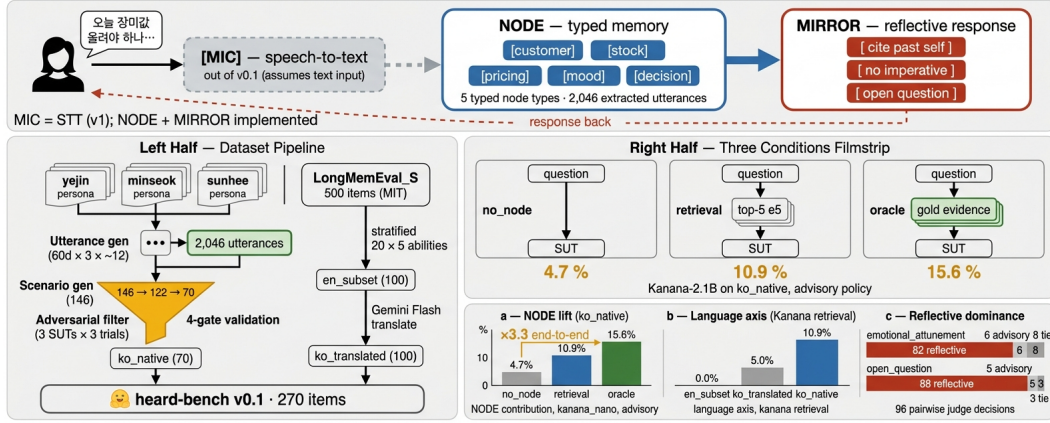


Figure 1: Heard v0.1 Pipeline & Headline Result Overview

Figure 1: Heard pipeline and headline results. Top band: the three pillars. Middle band: the dataset-construction funnel and the three-condition filmstrip (no_node 4.7%, retrieval 10.9%, oracle 15.6% for Kanana-2.1B). Bottom band: NODE lift, language-axis decay, and reflective-vs-advisory judge outcomes.

with. We ask two empirical questions: (1) does a lightweight retrieval-based memory pipeline yield measurable gains for a small Korean on-device LM on nightly self-talk questions; and (2) does a reflective prompt policy beat a generic advisory policy when memory is available?

Answering these requires a long-term-memory benchmark that is (a) Korean, (b) monologue (not dialogue), and (c) grounded in solo-business decisions. No existing benchmark satisfies all three (Table 1), so we construct heard-bench, a 270-item benchmark across three tracks, released on HuggingFace under CC-BY-4.0.

Table 1: No existing long-term-memory benchmark is simultaneously Korean, monologue, and solo-business; heard-bench fills the gap.

Benchmark	Language	Format	Domain
LongMemEval [14]	EN	user–AI dialogue	generic chat
PerLTQA [4]	ZH/EN	human–human dialogue	social events
LoCoMo [8]	EN	long dialogue	generic personal
heard-bench (ours)	KO	monologue	solo-business

1.2 Contributions

- Dataset as necessary infrastructure (§3.1).** We construct, adversarially filter, four-gate auto-validate, and author-review heard-bench (270 items across three tracks), released under CC-BY-4.0.
- Empirical NODE lift (§4.1).** Dense retrieval lifts the on-device target Kanana-2.1B from 4.7% to 10.9% to 15.6% on ko_native (3.3× end-to-end) and lifts every configuration in the sweep.
- Language-axis decay confirms Korean-native necessity (§4.3).** The ordering $\text{en_subset} \leq \text{ko_translated} < \text{ko_native}$ holds for ten of eleven SUTs.
- Reflective policy dominates (§4.4).** Across 528 pairwise decisions per rubric, reflective beats advisory on every rubric (88.1%/88.3% on the two headline rubrics).
- A v0.3 validation pass that closes its own limitations (§4.5–4.7).** We audit the judge for position bias (headline rubrics position-stable, non-directive unreliable), add the missing BM25 baseline (equal gold recall, scale- dependent reader use), and run the answer-first guardrail, which *fails* to close the $\text{retrieval} > \text{oracle}$ regression. Every v0.3 number reuses the v0.2 contains-token criterion and is traceable to experiments/20260612_v0.3_tier1/.

2 Related Work

Table 2 maps the works most directly built upon in the v0.3 pass to the upgrades they motivate.

Long-term memory benchmarks and agents. LongMemEval [14] introduces a five-ability taxonomy for chat-assistant memory (information extraction, multi-session reasoning, temporal reasoning, knowledge update, abstention); we adopt five of these and add a Heard-specific sixth, REFL. Lo-CoMo [8] builds long persona-grounded dialogues through a machine–human pipeline, the closest precedent for our synthetic corpus, and PerLTQA [4] motivates a typed-memory store. More recent memory layers—Mem0’s extraction/update pipeline [3], A-Mem’s self-organizing note graph [15], and the reflection-synthesis of generative agents [11]—offer design alternatives to NODE’s fixed typed schema; we situate NODE as the lightest of these, appropriate to an on-device budget. All of these benchmarks are English and dialogue-based, which is precisely the gap heard-bench fills. **On-device small LLMs, Korean models, and retrieval.** MiniRAG [5] shows that carefully engineered retrieval lets small models approach larger-model RAG quality, supporting our premise that the on-device constraint is met by the retriever, not the generator. Our retriever is multilingual-E5-small [13]; BGE-M3 [2] is the natural upgrade path. For generators we evaluate two Korean-native families, Kakao Kanana [1] and NAVER HyperCLOVA-X SEED [16], against the multilingual Qwen 2.5 [6] baseline and two Korean Llama-3 fine-tunes; the sparse baseline uses BM25 Okapi [12]. **Reflective dialogue and LLM-as-a-judge.** The MIRROR thesis draws on motivational interviewing; KMI [7] provides a Korean MI dataset whose behaviour taxonomy maps onto our four rubrics. Our pairwise evaluation follows MT-Bench/Chatbot Arena [17], which documents position bias; we mitigate it with A/B-swap and two judges, and in §4.6 audit the residual bias directly. **Quantization of small/multilingual LLMs.** Recent work shows automatic metrics hide degradation that judge/human evaluation reveals and that non-Latin-script languages are among the worst-affected [9], with 4-bit methods losing most on long-context, non-English inputs [10]; this frames our finding (§4.2) that the apparent int4 gains on the smallest Korean SUTs reduce to one borderline pair and one length artifact.

Table 2: Related work mapped to the v0.3 upgrades.

Cited work	Motivating limitation (§)	v0.3 experiment	Delta / conclusion
Zheng et al. 2023 (pairwise-judge position bias)	v0.2 pairwise wins reported without a position-bias audit	Task 1: order-consistency + inter-judge κ	emo/open robust (0.83/0.82, κ 0.42/0.49); non-directive collapses (0.34, κ 0.02)
Robertson & Zaragoza 2009 (BM25 Okapi)	§5.2: “BM25 baseline absent”	Task 2: BM25 vs dense, anchor SUTs	gold-hit@5 equal (38.9% = 38.9%); dense–BM25 +7.8 pp on 8 B, –3.1 pp on 2.1 B $\Rightarrow$ reader is the bottleneck
v0.2 §5.1 self-prediction (answer-first)	predicted answer-first closes retrieval > oracle	Task 3: advisory_answerfirst on oracle, 4 SUTs	refuted: 2/4 improve, none reach retrieval, mean gap-closed –5.8%; overlap ~unchanged

3 Methods

All code and artifacts are released (<https://github.com/cykim05/heard>, Apache-2.0); the dataset is at <https://huggingface.co/datasets/chanyoungkim/heard-bench> (CC-BY-4.0). A diskcache keyed on (model, messages, temperature, seed, response_format) makes a cold rerun free. MIC (a sounddevice recorder plus a faster-whisper transcriber, Korean fixed) ships in v0.1 but is not evaluated: all evaluation below operates on gold utterance text so the memory-pipeline question is isolated from STT error.

3.1 heard-bench construction

Personas and corpus. Three synthetic Korean solo-business owners anchor the corpus—a florist (Mapo), a roaster-café owner (Seongsu), and a one-chair hair-salon owner (Hongje)—each with regulars, stock or services, recurring stressors, and timestamped historical anchors. One generator call per (persona, day) block over 60 days produces ~ 12 utterances/day, rotating across Claude

Haiku 4.5, GPT-4o-mini, and Gemini 2.5 Flash so no single fingerprint dominates: 2,046 utterances (686/687/673), 43% referencing a historical anchor (the retrieval gold hooks).

Tracks and abilities. heard-bench has three tracks (Table 3); the native track is stratified by six abilities —IE (19), TR (15), KU (13), ABS (10), MR (7), and the Heard-specific REFL (6). IE, MR, KU, TR, ABS follow LongMemEval’s taxonomy.

Validation. ko_native candidates pass four stages. (1) *Adversarial filtering*: 150 generated, 146 parsed, 122 retained—keeping only items on which three no-memory baselines (Kanana-2.1B, Qwen-1.5B, Qwen-3B) fail across three trials each, so memory is structurally necessary. (2) *Four-gate auto-validation*: evidence–answer consistency, history-only fail reproduction, cross-generator 4-gram overlap $< 15\%$, ambiguity $\geq 3/5$. (3) *Author review* for naturalness and gold correctness. (4) *Final count*: 70 items; the dominant rejection cause at the clarity gate was rater conservatism on isolated questions (§5.2).

Table 3: The three heard-bench tracks (270 items total).

Track	Items	Language	Haystack	Source / License
en_subset	100	EN	EN	LongMemEval_S stratified / MIT
ko_translated	100	KO Q+ans	EN (kept)	Gemini 2.5 Flash translation / CC-BY-4.0
ko_native	70	KO	KO (ours)	persona-driven + adversarial filter / CC-BY-4.0

3.2 Experimental setup

Retrieval. The dense retriever is `intfloat/multilingual-e5-small` (118 M) [13]; mean-pooled, L2-normalized, cosine top-5, indexed per-persona for ko_native. The v0.3 sparse baseline (§4.5) swaps in BM25 Okapi [12] (`rank_bm25`) with `kiwipiepy` morpheme tokenization, leaving the rest unchanged.

Systems under test. The sweep evaluates eleven SUT configurations (Table 4) across four families and three scales, with a 4-bit NF4 variant wherever the model fits a 30 GB budget, all run sequentially on a single NVIDIA L40S 48 GB with embedder and SUT co-resident. Three further SUTs (EXAONE-3.5 2.4B, EEEV 10.8B, SOLAR 10.7B) were dropped for an import incompatibility or insufficient throughput.

Policies, conditions, judging. The *advisory* policy is a generic-assistant baseline (imperatives allowed, 3–8 sentences); the *reflective* policy must cite a retrieved memory verbatim, avoid imperatives, and end with an open question (3–6 sentences); the v0.3 *advisory_answerfirst* variant prepends one guardrail line. Three conditions: `no_node` (question only), `retrieval` (top-5), and `oracle` (all evidence documents, capped at $k \times 3 = 15$); decoding is `max_tokens=200`, `temperature=0.3`, `seed=42`. Reflective-quality pairs are scored by two judges (GPT-4o-mini, Claude Haiku 4.5) on four rubrics with A/B swap; the sweep yields 132 valid triples and 2,112 verdicts (528 per rubric) over 14,850 generations.

Table 4: SUT lineup. Korean-native families and the Korean-tuned Llama-3 derivatives populate the top of the ko_native results; the multilingual Qwen baselines never lead.

Logical id	Family	Params	Quant	License
kanana_nano	Kakao Kanana 1.5	2.09 B	fp16	Apache-2.0
kanana_nano_int4	Kakao Kanana 1.5	2.09 B	int4	Apache-2.0
kanana_8b	Kakao Kanana 1.5	8.03 B	fp16	Apache-2.0
kanana_8b_int4	Kakao Kanana 1.5	8.03 B	int4	Apache-2.0
hclova_seed_15b	NAVER HCX-SEED	1.59 B	fp16	HCX-SEED-Public
hclova_seed_15b_int4	NAVER HCX-SEED	1.59 B	int4	HCX-SEED-Public
qwen25_3b	Alibaba Qwen 2.5	3.09 B	fp16	Apache-2.0
qwen25_3b_int4	Alibaba Qwen 2.5	3.09 B	int4	Apache-2.0
qwen25_7b	Alibaba Qwen 2.5	7.62 B	fp16	Apache-2.0
bllossom_8b	Llama-3 (Korean FT)	8.03 B	fp16	Llama-3 Comm.
open_ko_8b	Llama-3 (Korean FT)	8.03 B	fp16	Llama-3 Comm.

4 Results

We report the v0.2 sweep (§4.1–4.4) and the v0.3 validation pass (§4.5–4.7). Figure 1 summarizes the architecture and headline trajectory; Figure 2 renders the three primary v0.2 findings with Wilson 95% CIs.

4.1 NODE lift across the lineup

Dense retrieval lifts contains-pass on `ko_native` for every SUT (Table 5): mean `retrieval` – `no_node` delta +11.8 pp, minimum +6.2 pp, no regressions. Korean-native and Korean-tuned families populate the top half; the multilingual Qwen baselines never lead. The on-device target `kanana_nano` follows the canonical 4.7% → 10.9% → 15.6% trajectory (3.3× end-to-end). A second pattern: `retrieval` and `oracle` no longer move in lockstep—five of eleven configurations record oracle strictly below retrieval, concentrated in the 7–8 B tier (e.g. `kanana_8b`: 7.8% vs. 18.8%) and absent in the smallest SUTs. We revisit this in §5.1.

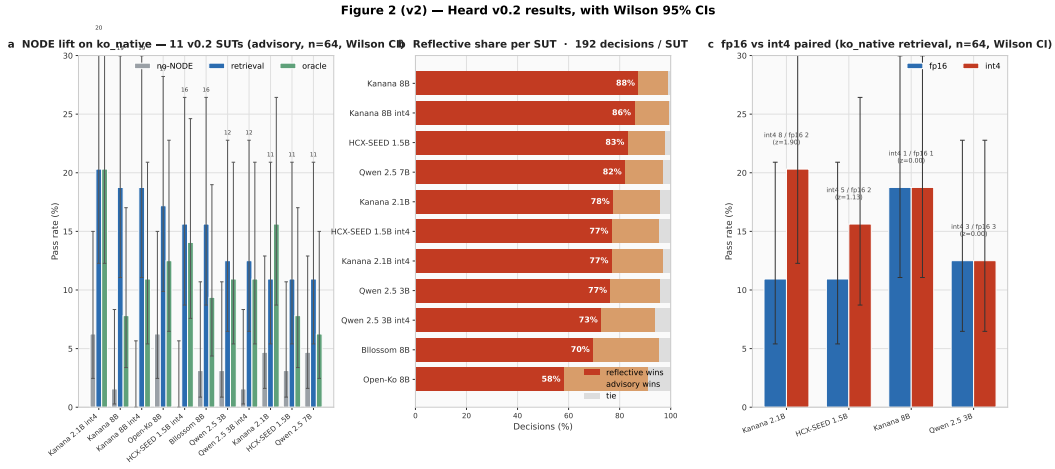


Figure 2: v0.2 results with Wilson 95% CIs. (a) NODE lift on `ko_native` (advisory, $n=64$), sorted by retrieval. (b) Reflective win-share per SUT (192 decisions/SUT). (c) fp16 vs. int4 paired on `ko_native` retrieval ($n=64$) with McNemar counts/ z . Colours: `no_node`=grey, `retrieval`=blue, `oracle`=sage, `advisory`=tan, `reflective`=red.

Table 5: Advisory pass rate on `ko_native` (contains-token, $n=64$), sorted by retrieval.

SUT	no_node	retrieval	oracle	Δ ret–no_node
kanana_nano_int4	6.2%	20.3%	20.3%	+14.1
kanana_8b	1.6%	18.8%	7.8%	+17.2
kanana_8b_int4	0.0%	18.8%	10.9%	+18.8
open_ko_8b	6.2%	17.2%	12.5%	+11.0
bllossom_8b	3.1%	15.6%	9.4%	+12.5
hclova_seed_15b_int4	0.0%	15.6%	14.1%	+15.6
qwen25_3b	3.1%	12.5%	10.9%	+9.4
qwen25_3b_int4	1.6%	12.5%	10.9%	+10.9
hclova_seed_15b	3.1%	10.9%	7.8%	+7.8
kanana_nano	4.7%	10.9%	15.6%	+6.2
qwen25_7b	4.7%	10.9%	6.2%	+6.2

4.2 Quantization, latency, and deployment recommendation

The marginal reading is that the two smallest SUTs gain under NF4 (`Kanana-2.1B` +9.4 pp, `HCLoVa-SEED-1.5B` +4.7 pp) while the larger two are unchanged (Table 6), but a paired item-level analysis modulates this. McNemar tests give `Kanana-2.1B` 8/2 ($z = 1.90$, $p \approx 0.06$), `HCLoVa-SEED-1.5B` 5/2

($z = 1.13$), Kanana-8B 1/1, Qwen-3B 3/3 ($z = 0$). Only Kanana-2.1B reaches marginal significance, on a 6-item swing at base rate $\approx 15\%$. The HCX-SEED gain is further confounded: its mean response length grows $261 \rightarrow 401$ characters (uniquely), and since contains-token is a substring metric, longer responses are mechanically more likely to contain gold tokens—so we treat that gain as a length artifact, consistent with the multilingual quantization literature [9, 10]. Latency is not improved by int4 here (three of four families $1.3\text{--}2.0\times$ slower; only `kanana_8b` neutral), so int4 is retained only as a memory option. Mean `ko_native` latency ranges 1.78–5.22 s, all within a 5–7 s decision-moment budget. Combining retrieval pass, reflective share, latency, and quantization sensitivity, we recommend `kanana_8b` as the default deployment SUT: second on retrieval (18.8%), first on reflective share (87.5%), mid-latency (3.73 s), and the one family where int4 is latency-neutral. `kanana_nano` remains right for the lightest envelope (1.83 s, 77.6%).

Table 6: int4 vs. fp16 on `ko_native` retrieval (advisory). Δ pass in pp; latency in mean s/generation ($n=64$).

SUT family	fp16	int4	Δ pass	fp16 lat.	int4 lat.
Kanana 2.1 B	10.9%	20.3%	+9.4	2.54	5.22
HCX-SEED 1.5 B	10.9%	15.6%	+4.7	1.83	3.45
Kanana 8.0 B	18.8%	18.8%	0.0	3.73	3.60
Qwen 3.0 B	12.5%	12.5%	0.0	2.66	4.81

4.3 Cross-track language axis

Mean advisory retrieval pass rates are 0.6% (`en_subset`), 2.9% (`ko_translated`), and 14.8% (`ko_native`)—a $25\times$ gap. The ordering `en_subset` $\leq$ `ko_translated` $<$ `ko_native` holds for ten of eleven SUTs (`open_ko_8b` ties the first two at 2.0%). The multilingual Qwen baselines do not close the `en_subset` gap despite English being native, which we attribute to LongMemEval haystack length and the absence of session-level retrieval. `ko_translated` sits closer to `en_subset` than to `ko_native` despite a Korean question and answer, confirming that haystack language dominates and motivating a full-haystack Korean track.

4.4 Reflective policy: accuracy-neutral, pairwise-dominant

On contains-token the reflective policy is approximately accuracy-neutral (mean advisory $\rightarrow$ reflective shift +0.9 pp, median 0.0, range -6.2 to $+12.5$): the hard constraints route the gold substring through a different stylistic shell rather than suppressing it. The pairwise judge measures the value of that shell (Table 7): reflective wins every rubric, with emotional attunement and open-question framing at 88.1% and 88.3% reflective share—recovering the v0.1 numbers under a $5.5\times$ -larger sample. Per-SUT reflective share spans 58.3% (`open_ko_8b`) to 87.5% (`kanana_8b`); the Llama-3 derivatives are weakest not because their reflective output is worse but because they follow advisory imperative templates more willingly, raising the baseline. §4.6 audits whether these margins survive a position-bias check.

Table 7: Pairwise judge totals per rubric (column sum = 528).

Rubric	Advisory	Reflective	Tie	Reflective share
specificity	158	348	22	65.9%
non_directive	178	350	0	66.3%
emotional_attunement	32	465	31	88.1%
open_question	33	466	29	88.3%

4.5 Sparse vs. dense retrieval

We add BM25 Okapi with kiwipiepy tokenization (Table 8); only the retriever changes, and an integrity check confirms that re-running dense retrieval reproduces the v0.2 figures exactly (`kanana_nano` 10.9%, `kanana_8b` 18.8%; top-5 ids matching on 64/64). Two observations separate retrieval quality from reader behaviour (Figure 3). First, the retrievers are equivalent at *finding* evidence: gold-hit@5

is identical at 38.9% (21/54). Second, they diverge *downstream*, and the direction flips with scale: on `kanana_8b` dense beats BM25 by +7.8 pp (18.8% vs. 10.9%), but on `kanana_nano` BM25 is ahead by 3.1 pp (14.1% vs. 10.9%). Because both surface gold evidence at the same rate, the gap is the reader’s *use* of an equally good context, not recall; at the 2.1 B on-device scale a far cheaper sparse retriever is not measurably worse.

Table 8: Sparse BM25 vs. dense retrieval on the two anchor SUTs; pass-rate cells include Wilson 95% CIs.

SUT	no_node	BM25	dense	$\Delta(\text{dense}-\text{BM25})$	gold-hit@5
<code>kanana_nano</code>	0.047 (0.016,0.129)	0.141 (0.076,0.246)	0.109 (0.054,0.209)	−0.031	0.389
<code>kanana_8b</code>	0.016 (0.003,0.083)	0.109 (0.054,0.209)	0.188 (0.111,0.300)	+0.078	0.389

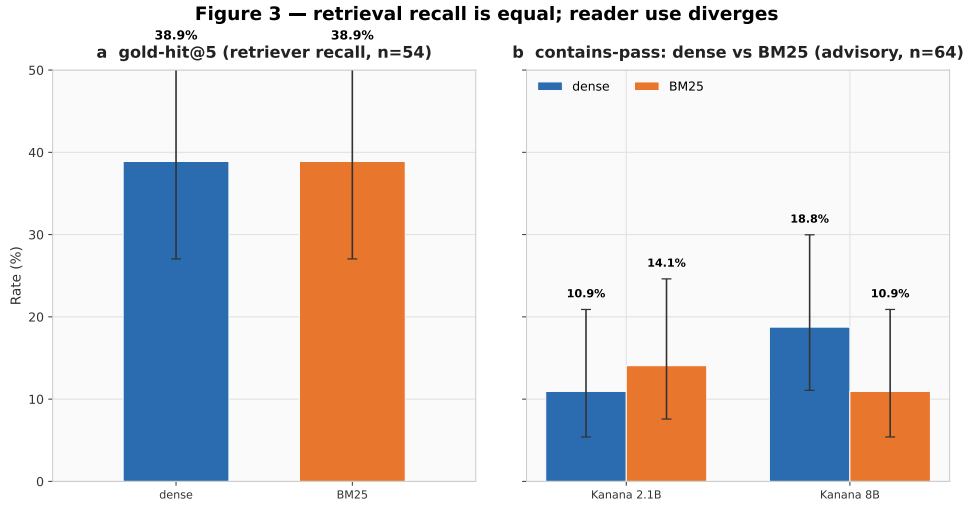


Figure 3: Retrieval recall is equal across retrievers; what differs is how the reader uses it. (a) gold-hit@5 near-identical for dense (38.9%) and BM25 (38.9%), $n=54$. (b) contains-token pass (advisory, $n=64$) for Kanana-2.1B and 8B; the dense advantage is SUT-dependent (+7.8 pp on 8B, −3.1 pp on 2.1B). Wilson 95% CIs. Takeaway: the bottleneck is the reader, not the retriever.

4.6 Judge reliability and position bias

We audit the existing 2,112 verdicts with no new judge calls. Because each (item, SUT, condition, rubric, judge) pair was scored in both A/B orders, we measure order-consistency and first-slot preference (0.5 is unbiased) and report inter-judge Cohen’s κ ; we do not report repetition stability, since identical calls hit the diskcache. The profile maps cleanly onto §4.4 (Table 9): the two headline rubrics are the most reliable (order-consistency 0.83/0.82, κ 0.42/0.49), while specificity (0.60, κ 0.21) and non-directive (0.34, κ 0.02, chance-level) are weak. The non-directive collapse is concentrated in one judge: GPT-4o-mini is order-consistent there only 1.5% of the time and picks the second slot 83% of the time, whereas Claude Haiku 4.5 is consistent 67%. Aggregate position bias is modest (first-slot share 0.385) and symmetrized away by A/B swap for the headline rubrics, so the 88% figures stand on the two position-stable rubrics, while the weaker margins should be read with the position artifact in mind.

4.7 The answer-first guardrail does not close the regression

§5.1 predicts that an “answer-first” guardrail would close most of the retrieval > oracle gap on the 7–8 B SUTs. We test it by prepending one line (`advisory_answerfirst`) and re-running only the oracle column on the four reversal SUTs (Table 10, Figure 4). The prediction is not confirmed: only two of four improve (`kanana_8b` +3.1 pp, `qwen25_7b` +1.6 pp), neither reaching retrieval level (closing 28.6%/33.3% of the gap), two regress, and the mean gap closed is −5.8%. The assumed mechanism is also absent: response length falls for the Kanana pair (216 → 181, 208 → 179

Table 9: Judge reliability and position bias by rubric.

Rubric	order-consistency	inter-judge κ	first-slot rate
specificity	0.602	0.210	0.451
non_directive [†]	0.341	0.016	0.174
emotional_attunement	0.826	0.420	0.451
open_question	0.818	0.493	0.475
Overall	0.647	0.257	0.385

[†]Reliability low for non_directive ($\kappa \approx 0.02$); interpretation deferred.

chars) yet the 4-gram recapitulation overlap is unchanged (e.g. 0.015 $\rightarrow$ 0.017) and even rises on `kanana_8b_int4` (0.016 $\rightarrow$ 0.025). Baseline overlap is already low (≈ 0.02), so the regression was never verbatim re-narration the guardrail could remove. We record the prediction as **refuted in its strong form**, weakly partial for `kanana_8b` alone.

Table 10: Answer-first guardrail on the four reversal SUTs (contains-token, $n=64$). gap-closed = fraction of the (retrieval – oracle_base) gap recovered.

SUT	oracle_base	oracle_af	retrieval	gap-closed	Δlen	overlap
<code>kanana_8b</code>	0.078	0.109	0.188	+28.6%	−34.5	0.015 $\rightarrow$ 0.017
<code>kanana_8b_int4</code>	0.109	0.063	0.188	−60.0%	−28.5	0.016 $\rightarrow$ 0.025
<code>qwen25_7b</code>	0.063	0.078	0.109	+33.3%	+8.1	0.020 $\rightarrow$ 0.018
<code>bllossom_8b</code>	0.094	0.078	0.156	−25.0%	−1.2	0.024 $\rightarrow$ 0.023

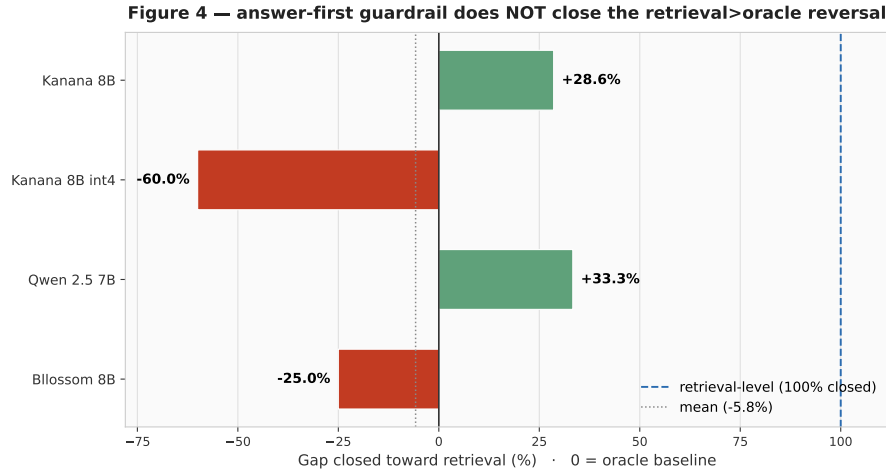


Figure 4: Answer-first guardrail does not close the `retrieval > oracle` reversal. Bars show the fraction of the gap closed per SUT; 0 is the oracle baseline, the dashed line at 100% marks retrieval-level. Two of four get worse, none reach retrieval, mean gap closed = −5.8%; the recapitulation proxy is unchanged, so the strong form is refuted.

5 Discussion

5.1 The reader bottleneck: the `retrieval > oracle` regression, revised

Absolute pass rates are low by construction—adversarial filtering retained only items where three no-memory baselines failed, so the headline interpretation is the relative lift ($3.3\times$), not the absolute rate. Within that, five of eleven configurations record oracle below retrieval, concentrated in the 7–8 B tier. We initially read this as a long-context summarization bias: a 15-document oracle prompt roughly triples the retrieval-prompt length, so larger SUTs might recapitulate rather than answer.

That account yields the falsifiable prediction tested in §4.7—and it fails (mean -5.8% gap closed, overlap unchanged even where length drops). We therefore now read the regression as a genuine context-utilization effect rather than a prompt artifact: larger SUTs extract the answer less reliably from a longer, cleaner gold context than from a shorter retrieval context. The §4.5 finding—that retrieval quality is not the bottleneck—points the same way, and the judge audit (§4.6) shows the reflective claim rests on the two position-stable rubrics, a more honest footing than raw win-shares. We recommend any LLM-judged reflective result report order-consistency and inter-judge κ alongside win-shares. Characterizing the utilization gap (position-in-prompt sensitivity, evidence-count scaling) is the open question the failed guardrail leaves behind.

5.2 Limitations and future work

The native track is 70 items rather than the planned 100 (rather conservatism on isolated questions, a one-line prompt fix), and absolute numbers rest on a 64-item denominator where a single configuration’s standard error is $\approx \pm 5$ pp, so we lean on cross-lineup consistency. The `ko_translated` haystack stays English, partially confounding the language axis, and heard-bench is synthetic, so a human-validated subset would strengthen ecological validity. The benchmark scores substring recall, which leaves three NODE-relevant axes unmeasured: absence queries (cosine similarity always returns a neighbour), entity aggregation (needs traversal, not bounded top- k), and temporal-latest queries (needs timestamp filtering); a NODE-native benchmark of 30–50 items targeting these is the natural extension. Finally, MIC is unevaluated, a BM25-vs-dense comparison beyond the two anchor SUTs and a full-haystack Korean track remain open, and the dropped 10 B+ Korean SUTs are future candidates.

6 Conclusion

heard-bench is, to our knowledge, the first Korean long-term-memory benchmark targeting solo-business monologue. An eleven-configuration sweep establishes three robust findings: dense retrieval lifts contains-pass for every configuration (mean $+11.8$ pp), a reflective policy dominates advisory on every rubric (88.1%/88.3% on the two “mirror, do not advise” rubrics), and a strict language-axis ordering favours Korean-native data. A v0.3 validation pass then closes three of the report’s own limitations at zero paid-API cost: a position-bias audit confirms the reflective claim rests on the two position-stable rubrics (κ 0.42/0.49); a BM25 baseline matches dense gold recall (38.9%) while diverging by scale, locating the bottleneck in the reader; and the answer-first guardrail is refuted in its strong form, reclassifying the `retrieval > oracle` regression from a prompt artifact to a genuine context-utilization gap. We recommend `kanana_8b` as the default deployment SUT and identify reader-side context utilization—not retrieval recall—as the limiting factor for the next iteration.

Acknowledgments and AI use

Claude Code was used as a coding assistant for the pipeline, the v0.3 experiments, and figure-generation scripts, and AI tools were used for drafting assistance. All design decisions—research framing, dataset methodology, hypothesis formulation, result interpretation, and writing direction—were made by the author, who is responsible for the final content.

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